

AERODYNAMIC FORCES AND MOMENTS

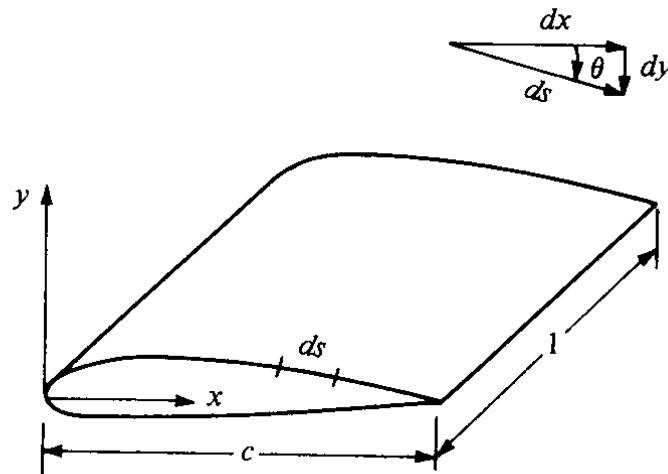


Figure 1.15 Geometrical relationship of differential lengths.

$$dx = ds \cos \theta$$

$$dy = -(ds \sin \theta)$$

$$S = c(1)$$

$$C_l = C_n \cos \alpha - C_a \sin \alpha$$

$$C_d = C_n \sin \alpha + C_a \cos \alpha$$

$$C_n = \frac{1}{c} \left[\int_0^c (C_{p,l} - C_{p,u}) dx + \int_0^c \left(c_{f,u} \frac{dy_u}{dx} + c_{f,l} \frac{dy_l}{dx} \right) dx \right]$$

$$C_a = \frac{1}{c} \left[\int_0^c \left(C_{p,u} \frac{dy_u}{dx} - C_{p,l} \frac{dy_l}{dx} \right) dx + \int_0^c (c_{f,u} + c_{f,l}) dx \right]$$

$$C_{m_{LE}} = \frac{1}{c^2} \left[\int_0^c (C_{p,u} - C_{p,l}) x dx + \int_0^c \left(c_{f,u} \frac{dy_u}{dx} + c_{f,l} \frac{dy_l}{dx} \right) x dx \right. \\ \left. + \int_0^c \left(C_{p,u} \frac{dy_u}{dx} + c_{f,u} \right) y_u dx + \int_0^c \left(-C_{p,l} \frac{dy_l}{dx} + c_{f,l} \right) y_l dx \right]$$